

Name: Solutions

Student ID: _____

1. (8 points) Expand the following logarithmic functions as much as possible:

(a) $\ln(5x^2y)$

(b) $\ln(\sqrt{xy}/z^3)$

(c) $\ln(6x\sqrt{y^2 + y/z})$

(d) $\ln(e^{2x}y^3)$

$$(a) \ln(5) + \ln(x^2) + \ln(y) = \boxed{\ln(5) + 2\ln(x) + \ln(y)}$$

$$\begin{aligned} (b) \ln(\sqrt{xy}) - \ln(z^3) &= \ln((xy)^{\frac{1}{2}}) - 3\ln(z) \\ &= \frac{1}{2}\ln(xy) - 3\ln(z) \\ &= \boxed{\frac{1}{2}(\ln(x) + \ln(y)) - 3\ln(z)} \end{aligned}$$

$$\begin{aligned} (c) \ln(6x) + \ln(\sqrt{y^2 + y/z}) &= \ln(6) + \ln(x) + \ln((y^2 + y/z)^{\frac{1}{2}}) \\ &= \boxed{\ln(6) + \ln(x) + \frac{1}{2}\ln(y^2 + y/z)} \end{aligned}$$

$$\begin{aligned} (d) \ln(e^{2x}) + \ln(y^3) &= 2x \ln(e) + 3\ln(y) \\ &= \boxed{2x + 3\ln(y)} \end{aligned}$$

2. (8 points) Take the first partial derivatives of the following functions **with respect to y**:

(a) $f(x, y) = \ln(x^5) + y^2/x$

(b) $f(x, y) = e^{x^{1/3}y^{1/3}}$

(c) $f(x, y) = 3x^2y - 2xy^3 + 4y$

(d) $f(x, y) = \ln(x^2 + y^2)$

$$(a) f_y = 0 + \frac{2y}{x} = \boxed{2\frac{y}{x}}$$

$$(b) f_y = e^{x^{1/3}y^{1/3}} (x^{1/3} (\frac{1}{3}) y^{-2/3})$$

$$= \boxed{e^{x^{1/3}y^{1/3}} \left(\frac{x^{1/3}}{3y^{2/3}} \right)}$$

$$(c) f_y = 3x^2(1) - 2x(3y^2) + 4(1)$$

$$= \boxed{3x^2 - 6xy^2 + 4}$$

$$(d) f_y = \frac{1}{x^2 + y^2} (2y) = \boxed{\frac{2y}{x^2 + y^2}}$$

3. (12 points) Find the critical points (if any) for the following function. For each critical point determine whether the function is at a local minimum or maximum.

$$f(x) = 5xe^{-x}$$

Foc

$$f'(x) = 5(1)e^{-x} + 5x(-1)e^{-x} = 0$$

$$5e^{-x} - 5xe^{-x} = 0$$

$$\cancel{5e^{-x}} = \cancel{5xe^{-x}}$$

$$\boxed{x^* = 1}$$

* Product rule

$$* \frac{d(e^{-x})}{dx} = -e^{-x}$$

Soc

$$f''(x) = -5e^{-x} - (5e^{-x} + (-1)5xe^{-x})$$

$$= -5e^{-x} - 5e^{-x} + 5xe^{-x}$$

$$= -10e^{-x} + 5xe^{-x}$$

$$f''(x^* = 1) = -10e^{-1} + 5(1)e^{-1}$$

$$= -5e^{-1} = -\frac{5}{e} < 0$$

Since $e \approx 2.7 > 0$

$$\Rightarrow \boxed{\text{maximum}}$$

4. (12 points) Consider the function $f(x, y) = xy$. Find the values x^* and y^* that maximize f when x and y must sum to 1. (Hint: use Lagrange and think of 1 as income in a budget constraint.)

$$\text{Constraint : } x + y = 1 \Rightarrow 1 - x - y = 0$$

$$\mathcal{L} = xy + \lambda (1 - x - y)$$

Foc

$$\begin{array}{l} \textcircled{1} y - \lambda = 0 \\ \textcircled{2} x - \lambda = 0 \end{array} \left. \vphantom{\begin{array}{l} \textcircled{1} y - \lambda = 0 \\ \textcircled{2} x - \lambda = 0 \end{array}} \right\} \begin{array}{l} y = \lambda \\ x = \lambda \end{array} \Rightarrow x = y$$

$$\textcircled{3} 1 = x + y$$

$$1 = x + x$$

$$1 = 2x$$

$$\boxed{x^* = \frac{1}{2} \quad y^* = \frac{1}{2}}$$

Bonus: Name an economist who won a Nobel Prize this week.

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